

Artificial Intelligence in the Knowledge Economy

Capability, autonomy, and the price of expertise

davidolano03

Ide & Talamàs (2025), Journal of Political Economy

github.com/davidolano03/ai-05-ide

Course Repository 5

Source read: arXiv v11, title page dated February 25, 2025 (35 pages)

The economic question

Knowledge can be copied into AI agents. Human time cannot.

Capability: what does AI know?

An agent with knowledge a solves a fraction a of problems.

Autonomy: what may AI do?

Autonomous AI can produce and advise.
A co-pilot can only advise.

Who gains at the bottom, who gains at the top, and how much does the economy produce?

The comparison is between equilibria: occupations, matches, and wages all adjust.

Source: arXiv v11, Sections 3 and 6.

A firm allocates scarce problem-solving time

One worker has knowledge z and encounters $x \sim U[0, 1]$. A solver knows $s > z$.

$$\underbrace{n(z)}_{\text{workers}} \underbrace{(1-z)}_{\text{questions per worker}} \underbrace{h}_{\text{time per question}} = 1 \implies n(z) = \frac{1}{h(1-z)}.$$

$$\Pi = n(z) [s - w(z)] - w(s).$$

People maximize income. Firms choose their structure and hires; competition gives zero profits and market clearing.

More worker knowledge saves solver time. More solver knowledge helps every worker in the team.

Source: arXiv v11, pp. 10–13. Each question costs time even if the solver cannot answer.

Conditions behind the main results

- Unit mass of humans, $z \in [0, 1]$, with continuous, strictly positive density g .
- Exogenous, observable knowledge; one unit of time per person; risk neutrality.
- Competitive firms, no fixed costs, at most two layers, abundant opportunities.
- $0 < h < h_0(G) < 1$: no independent human producers before AI.
- One AI capability $a \in [0, 1)$, common access, one agent per unit of compute.
- Compute is abundant relative to human time. A sufficient condition is

$$\mu > \int_0^a h(1-z) dG(z) + \frac{1-G(a)}{h(1-a)}.$$

- Autonomy comparisons hold G, h, a, μ fixed and retain the two-layer limit.

Source: arXiv v11, pp. 9–14, 18 and 26. In particular, $a = 1$ is excluded.

Proposition 5: bottom gains require enough capability

Write w for pre-AI wages and w^* for autonomous-AI wages. Define

$$B = \{z \in [0, a] : w^*(z) > w(z)\}, \quad T = \{z \in [a, 1] : w^*(z) > w(z)\}.$$

Under the baseline conditions

$$B \neq \emptyset \iff a > \bar{a}, \quad \bar{a} \in \text{int } W.$$

$$T \neq \emptyset \quad \text{for every } a \in [0, 1).$$

Here W is the set of workers *before* AI. Winners occupy tails of the distribution; the human type $z = a$ loses.

Source: arXiv v11, pp. 24–25. Top gains depend on $h < h_0$ and $a < 1$.

Proposition 6: the co-pilot threshold is $w(0)$

Let w^* denote wages with non-autonomous AI.

$$a \leq w(0)$$

No AI adoption in equilibrium. Human occupations and wages stay at their pre-AI values.

$$a > w(0)$$

The least knowledgeable use AI. Some people strictly gain, and some strictly lose.

The equilibrium is unique and efficient. It maximizes labor income among feasible co-pilot allocations.

$$r^* = 0 \quad \text{because some compute is idle.}$$

The pre-AI wage $w(0)$ includes the value of organizational support. A co-pilot must beat that outside option.

Source: arXiv v11, pp. 26–27; AI users lie below human-assisted workers, independent producers, and solvers.

Proposition 6: comparisons and strictness

Under the same baseline conditions:

$$\begin{aligned} Y^* &> Y^*, \\ \exists z \in (0, 1] : \quad w^*(z) &\leq w(z), \\ \exists \varepsilon_b > 0 \, \forall z \in [0, \varepsilon_b) : \quad w^*(z) &\geq \max\{w(z), w^*(z)\}, \\ \exists \varepsilon_t > 0 \, \forall z \in (1 - \varepsilon_t, 1] : \quad w^*(z) &\leq w^*(z). \end{aligned}$$

The second and third inequalities are strict if $a > w(0)$. The last is strict for $z \neq 1$; the endpoint only has a weak claim.

These are local tail comparisons, not rankings for every worker.

Source: arXiv v11, Proposition 6, p. 27. Y means total output, not labor income.

Why autonomy changes the opportunity cost of advice

Autonomous AI

Spare compute can produce, so $r^* = a$.

$$w^*(z) = a - h(1 - z)a$$

for a human worker using AI advice.

$$w^*(s) = \frac{s - a}{h(1 - a)}$$

for a human solver supervising AI.

Non-autonomous AI

Spare compute is idle, so $r^* = 0$.

$$w^*(z) = a$$

for a human worker using AI advice.
AI cannot supply routine workers to human experts. It can compete with experts as a solver.

Source: arXiv v11, pp. 19–20 and 26–27. These formulas apply to the stated active occupations.

A three-type economy we can solve by hand

$$(z_L, z_M, z_H) = (0, \frac{1}{2}, 1), \quad (m_L, m_M, m_H) = (\frac{1}{2}, \frac{3}{10}, \frac{1}{5}), \quad h = \frac{1}{2}.$$

Let x_{ij} measure workers of type i assisted by type j . An allocation $(x_{LM}, x_{LH}, x_{MH}) = (\frac{1}{5}, \frac{3}{10}, \frac{1}{5})$ uses all resources.

$$w_L + \frac{1}{2}w_M = \frac{1}{2}, \quad w_L + \frac{1}{2}w_H = 1, \quad w_M + \frac{1}{4}w_H = 1.$$

$$\boxed{(w_L, w_M, w_H) = (\frac{1}{5}, \frac{3}{5}, \frac{8}{5})}, \quad Y = \frac{3}{5}.$$

Wages exceed independent output, so unused single-person firms cannot profit.

Own finite model: divisible masses, three knowledge atoms. This changes the paper's domain.

The capability threshold comes from one subtraction

Autonomous AI, $\mu = 10$, $0 \leq a \leq 1/3$. Low types work with middle types; remaining middle and high types supervise AI.

$$w_H^* = 2, \quad w_M^* = \frac{1 - 2a}{1 - a}, \quad w_L^* = \frac{1}{2} - \frac{1}{2}w_M^* = \frac{a}{2(1 - a)}.$$

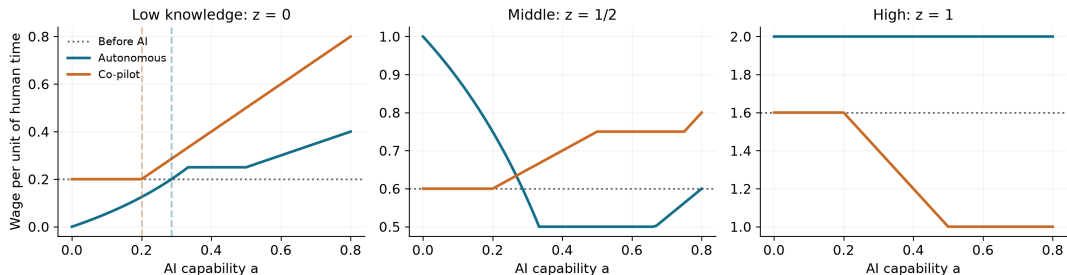
The sign of the gain

$$w_L^* - \frac{1}{5} = \frac{7a - 2}{10(1 - a)} \implies w_L^* > \frac{1}{5} \iff a > \frac{2}{7}.$$

At $a = 2/7$ there is equality. At $a = 1/10$ the low wage falls to $1/18$; at $a = 3/10$ it rises to $3/14$.

Autonomy is unchanged. Capability changes the sign.

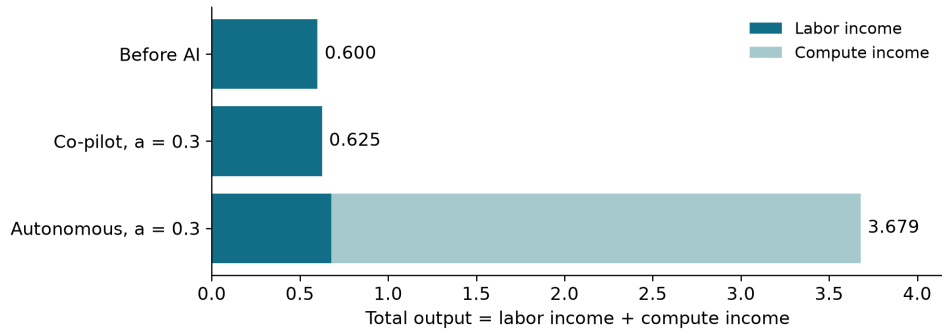
Same people, different AI roles



Own three-type model, $h = 1/2$, $\mu = 10$. Dotted lines: wages before AI.

The low-type thresholds are $1/5$ for co-pilots and $2/7$ for autonomy. Low types can gain from autonomy while still using human supervisors.

More output does not mean everyone earns more



At $a = 0.3$, autonomous labor income is $19/28 \approx 0.679$; total output is $103/28 \approx 3.679$.

The difference includes $\mu r = 3$ of compute income. Co-pilot output equals labor income because its compute rent is zero.

What the computation checks

$$\max_{x \geq 0} c^\top x \quad \text{s.t. } Ax \leq b \quad \longleftrightarrow \quad \min_{p \geq 0} b^\top p \quad \text{s.t. } A^\top p \geq c.$$

Columns are feasible firm activities. Resources are human types and compute. Dual prices are wages and the compute rent.

- Exact symbolic check of the hand algebra with SymPy.
- 166 equilibria: feasibility, zero duality gap, and complementarity at 10^{-8} tolerance.
- Wage ranges on the optimal dual face detect non-unique prices.

A two-type control admits multiple wages. Another violates top gains after changing the paper's domain. Neither is a refutation of its continuum theorem.

Lean: the paper's wage identity

For an active firm with an AI solver (Proposition 2, pp. 19–20):

$$n(a - w) - r = 0, \quad r = a, \quad n^{-1} = h(1 - z) \implies w = a[1 - h(1 - z)].$$

```
theorem automatedWorkerWage_of_zeroProfit
  {n a w r h z : ℝ} (hn : n ≠ 0)
  (hzero : n * (a - w) - r = 0) (hrent : r = a)
  (hteam : 1 / n = h * (1 - z)) :
  w = a * (1 - h * (1 - z)) := by
```

```
have hsub : a - w = a / n := by
  apply (eq_div_iff hn).2
  nlinarith [hmul]
```

Selected proof step: `hmul` is $n(a - w) = a$. `eq_div_iff hn` justifies division by nonzero n ; `nlinarith` checks the algebra. The real-valued premises supply zero profit, rent, and team size. Lean does not derive their equilibrium origin here.

`MainTheorems.lean`: direct Lean check passed (exit 0). Full-paper check reported next.

Lean: capability determines the sign in our example

Own finite wage formula, with the economic domain stated:

$$0 \leq a \leq \frac{1}{3} : \quad \frac{1}{5} < \frac{a}{2(1-a)} \iff \frac{2}{7} < a.$$

```
theorem autonomousBottomWage_gt_baseline_iff
  {a : ℝ} (ha0 : 0 ≤ a) (ha13 : a ≤ 1 / 3) :
  baselineBottomWage < autonomousBottomWage a ↔ (2 / 7 : ℝ) < a := by
```

```
have hden : 0 < 2 * (1 - a) := by
  norm_num at ha13 ⊢
  nlinarith
```

`hden` proves the denominator positive. The proof then uses `lt_div_iff₀` to multiply without reversing the inequality and checks both directions. Lean checks the comparison of defined wages; hand calculations and the LP check their equilibrium interpretation.

Paper build: exit 0 with two sorry warnings. Workflow check: interrupted after about 90 s, not passed.

Continuum translation and proofs remain incomplete.

Handwritten check: time and resource constraints

Handwritten mathematical work on grid paper:

$h(1-z)n(z) = 1 \rightarrow$ restricción de tiempo del supervisor

$n(z) = \frac{1}{h(1-z)}$ función del equipo

$h = 1/2 \rightarrow n(0) = 2 \quad n(1/2) = 4$
 $z_L \quad z_M$

Eg. sin IA:

$x_{LM} = 1/5 ; x_{LH} = 3/10 ; x_{MH} = 1/5$

$L: 1/5 + 3/10 = 1/2$

$M: 1/2 \cdot 1/5 + 1/5 = 3/10$

$H: 1/2 \cdot 3/10 + 1/4 \cdot 1/5 = 1/5$

The handwritten calculation checks solver time:

$$h(1-z)n(z) = 1.$$

For $h = 1/2$, a solver can assist two low-type workers or four middle-type workers.

The proposed allocation uses every type's available time. Zero profit then gives

$$(w_L, w_M, w_H) = \left(\frac{1}{5}, \frac{3}{5}, \frac{8}{5}\right).$$

Original student photographs appear in full in the appendix.

Handwritten check: the threshold and the verdict

Autonomous AI, $\mu = 10$, $0 \leq a \leq 1/3$. The handwritten comparison is

$$w_L^* = \frac{a}{2(1-a)} > \frac{1}{5} \iff 7a > 2 \iff a > \frac{2}{7}.$$

¿cuándo ganan los trabajadores
bajos?

$$w_L^* > w_L \iff \frac{a}{2(1-a)} > 1/5 \quad \text{como } 0 \leq a \leq 1/3$$
$$\rightarrow 7a > 2$$
$$w_L^* > w_L \iff a > 2/7.$$

si $a = 2/7$ salario no cambia.

Since $2(1-a) > 0$, multiplication preserves the inequality.

At $a = 2/7$, the wage is unchanged.

Verdict: capability changes who gains even when autonomy is fixed. The value $2/7$ belongs to this finite example.

What the checks taught us

- Autonomy determines feasible roles and the opportunity cost of compute. Capability determines whether those roles improve a worker's outside option.
- A claim about gains needs a benchmark, a domain, and a strictness condition.
- A finite model makes the mechanism transparent, but its atoms can change equilibrium properties.
- A Lean proof checks its formal statement. Matching that statement to the paper remains a separate task.

The key correction: bottom winners under autonomy require $a > \bar{a}$; strict bottom gains with a co-pilot require $a > w(0)$.

Original handwritten derivation: both pages

ALINA INTERNATIONAL SAC
CARE YOUR LIFE

FECHA:

Derivación de un umbral de capacidad para las ganancias de los trabajadores menos informados

3 tipos de personas:

$(z_L, z_M, z_H) = (0, 1/2, 1)$ *capacidades*

$(m_L, m_M, m_H) = (1/2, 3/10, 1/5)$ *masas*

costo: $h = 1/2$ y persona tiene 1 unidad de tiempo.

$z \rightarrow$ nivel de conocimiento y fracción de problemas que resuelve tecnológicamente.

$1-z \rightarrow$ fracción restante a resolver.

$h(1-z)n(z) = 1 \rightarrow$ restricción de tiempo del supervisor

$n(z) = \frac{1}{h(1-z)}$ *inverso del tiempo*

$h = 1/2 \rightarrow n(0) = 2 \quad n(1/2) = 4$

Eg. sin IA:

$x_{LM} = 1/5 ; x_{MH} = 3/10 ; x_{NH} = 1/5$

$L : 1/5 + 3/10 = 1/2$

$M : 1/2 \cdot 1/5 + 1/5 = 3/10$

$H : 1/2 \cdot 3/10 + 1/4 \cdot 1/5 = 1/5$

IN: LOS ALBARROBOS NRO. 560 URB. LOS JARDINES (PRIMER PISO) LIMA - LIMA - SAN MARTIN DE PORRES

ALINA INTERNATIONAL SAC
CARE YOUR LIFE

FECHA:

Beneficio cero: $w_L + h(1-z_L)w_H = z$

En act. implícita: $w_L + 1/4 w_M = 1/2$

$w_L + 1/2 w_H = 1$

$w_M + 1/4 w_H = 1 \rightarrow w_M = 3/5$

$w_H = 5/5$

$w_L = 1/5$

Con IA:

$0 \leq a \leq 1/3$ *capacidad* $\mu = 10$

$r = a$ *salarios supervisor productividad individual de c/ tipo.*

$w^*(s) = n(a)(s-r) = (s-a) \cdot 1/2(1-a)$

$w_M^* = 2$

$w_H^* = \frac{1-2a}{1-a}$

condición: $w_L^* + 1/2 w_M^* = 1/2$

$w_L^* = \frac{a}{2(1-a)}$

¿cuándo ganan los trabajadores bonos?

$w_L^* > w_L \Leftrightarrow \frac{a}{2(1-a)} > 1/5$ como $0 \leq a \leq 1/3$

$w_L^* > w_L \Leftrightarrow a > 2/7 \rightarrow \nexists a > 2$

si $a = 2/7$ salario no cambia.

IN: LOS ALBARROBOS NRO. 560 URB. LOS JARDINES (PRIMER PISO) LIMA - LIMA - SAN MARTIN DE PORRES

Original photographs supplied by the student. Handwriting preserved in Spanish.

Extension: output can rise while labor income falls

In our autonomous three-type economy, for $0 \leq a \leq 1/3$:

$$\mathcal{L}^*(a) = \frac{1}{2}w_L^* + \frac{3}{10}w_M^* + \frac{1}{5}w_H^* = \frac{14 - 15a}{20(1 - a)}.$$

$$\frac{d\mathcal{L}^*}{da} = -\frac{1}{20(1 - a)^2} < 0, \quad \frac{dY^*}{da} = \mu - \frac{1}{20(1 - a)^2} > 0 \quad (\mu = 10).$$

Low wages rise and high wages stay fixed. Falling middle wages dominate the change in aggregate labor income.

This compares *different autonomous capabilities*. Labor income still exceeds the no-AI baseline throughout this interval.

Own extension; exact symbolic verification. No claim is made outside this interval.

Source version and reproducibility

Ide, E., & Talamàs, E. (2025). *Artificial Intelligence in the Knowledge Economy*. JPE 133(12), 3762–3800.

doi:10.1086/737233

Read and pinned: arXiv:2312.05481v11, 35 pages, title page February 25, 2025. The deposit metadata says February 24. The course's May manuscript and arXiv v12 were not substituted.

analysis/: derivation and solver. results/: checks and data.

lean/: complete generated paper folder, preserving workflow artifacts.

LEAN_RUN.md: exact run, scope, build/check result and blockers.

github.com/davidolano03/ai-05-ide